

Soluciones Examen Final

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Problem 1 - Examen Final

- ① (12 points) In a photon-photon fusion experiment, two photons of energies E_1 and E_2 move in the opposite direction in the laboratory frame, collide with each other and produce two mesons M_1 and M_2 with unequal masses via the process $\gamma\gamma \rightarrow M_1 + M_2$. The meson M_1 emerges at angle α_1 with respect to collision axis.
- ① (4 points) Find the energies of the produced mesons M_1 , M_2 and their relative velocity v_{rel} .
 - ② (4 points) Assuming that photon energies are equal, $E_1 = E_2$, find the minimal photon energy E_e required for this process.
 - ③ (4 points) Write out explicit expressions for the Mandelstam variables s , t , u in terms of energies and the angle α in the lab frame.

Resembles a lot the problems from Certamen 1 ($ee \rightarrow \pi^+\pi^-$), just two minor differences:

- Photons are massless, $m_\gamma^2 = 0$; and electrons are not, $m_e^2 \neq 0$.
- Produced mesons M_1, M_2 now may have different masses

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Solution: In the lab frame, which coincides with center-of-mass frame in this problem, 4-momenta of γ_1, γ_2 are

$$k_1^\mu = (E_{k_1}/c, \mathbf{k}_1), \quad k_2^\mu = (E_{k_2}/c, \mathbf{k}_2), \quad E_{k_a} = c\sqrt{\mathbf{k}_a^2} \quad (1)$$

we assume that both $\mathbf{k}_1, \mathbf{k}_2$ are vectors in direction $\hat{\mathbf{z}}$.

–We'll use $c = 1$ from now on for brevity

–We'll start with **part 3** and introduce explicit results for Mandelstam variables (later we'll use them as shorthand notations):

$$s = (k_1^\mu + k_2^\mu)^2 = (E_{k_1} + E_{k_2}, (E_{k_1} - E_{k_2})\hat{\mathbf{z}})^2 = 4E_{k_1}E_{k_2},$$

$$\begin{aligned} t &= (k_1^\mu - p_{M_1}^\mu)^2 = k_1^2 + p_{M_1}^2 - 2k_1 \cdot p_{M_1} = M_1^2 - 2 \left(E_{k_1} \sqrt{\mathbf{p}_{M_1}^2 + M_1^2} - E_{k_1} |\mathbf{p}_{M_1}| \cos \alpha \right) \\ &= M_1^2 - 2 \left(E_{k_1} E_{M_1} - E_{k_1} \sqrt{E_{M_1}^2 - M_1^2} \cos \alpha \right) \end{aligned}$$

$$\begin{aligned} u &= (k_1^\mu - p_{M_2}^\mu)^2 = (\text{use 4-momentum conservation}) = (k_2^\mu - p_{M_1}^\mu)^2 \\ &= k_2^2 + p_{M_1}^2 - 2k_2 \cdot p_{M_1} = M_1^2 - 2 \left(E_{k_2} \sqrt{\mathbf{p}_{M_1}^2 + M_1^2} + E_{k_2} |\mathbf{p}_{M_1}| \cos \alpha \right) \\ &= M_1^2 - 2 \left(E_{k_2} E_{M_1} + E_{k_2} \sqrt{E_{M_1}^2 - M_1^2} \cos \alpha \right) \end{aligned}$$

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In the lab frame, which coincides with center-of-mass frame in this problem, 4-momenta of γ_1, γ_2 are

$$k_1^\mu = (E_{k_1}, \mathbf{k}_1), \quad k_2^\mu = (E_{k_2}, \mathbf{k}_2), \quad E_{k_a} = \sqrt{\mathbf{k}_a^2}, \quad a = 1, 2 \quad (1)$$

we assume that both $\mathbf{k}_1, \mathbf{k}_2$ are vectors in direction $\hat{\mathbf{z}}$.

► After interaction we produce meson pair M_1, M_2 , whose momenta in the center-of-mass-frame are

$$p_{M_1}^\mu = (E_{M_1}, \mathbf{p}_1), \quad p_{M_2}^\mu = (E_{M_2}, \mathbf{p}_2), \quad E_{M_a} = \sqrt{\mathbf{p}_a^2 + M_a^2}, \quad a = 1, 2 \quad (2)$$

In view of the energy-momentum conservation and geometrical considerations,

$$\mathcal{E} = E_{k_1} + E_{k_2} = E_{M_1} + E_{M_2},$$

$$p_{1z} + p_{2z} = \mathcal{P} \equiv k_{1z} - k_{2z} = E_{k_1} - E_{k_2}$$

$$\mathbf{p}_{1\perp} = -\mathbf{p}_{2\perp} \equiv \mathbf{p}_\perp, \quad p_{1z}/p_\perp = \cot \alpha$$

$$\Rightarrow p_{1z} = |p| \cos \alpha = \cos \alpha \sqrt{E_{M_1}^2 - M_1^2}, \quad |\mathbf{p}_\perp| = \sin \alpha \sqrt{E_{M_1}^2 - M_1^2}$$

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Energy conservation:

$$\begin{aligned}\Rightarrow \mathcal{E} &= \sqrt{p_{\perp}^2 + p_{1z}^2 + M_1^2} + \sqrt{p_{\perp}^2 + (\mathcal{P} - p_{1z})^2 + M_2^2} = \\ &= E_{M_1} + \sqrt{E_{M_1}^2 + M_2^2 - M_1^2 + \mathcal{P}^2 - 2\mathcal{P}p_{1z}} = \\ &= E_{M_1} + \sqrt{E_{M_1}^2 + M_2^2 - M_1^2 + \mathcal{P}^2 - 2\mathcal{P} \cos \alpha \sqrt{E_{M_1}^2 - M_1^2}}\end{aligned}$$

$$\begin{aligned}\Rightarrow E_1 &= \frac{\mathcal{E} (s + M_1^2 - M_2^2)}{2(\mathcal{E}^2 - \mathcal{P}^2 \cos^2(\alpha))} + \\ &+ \frac{\mathcal{P} \cos(\alpha) \sqrt{M_1^4 - 2M_1^2 (\mathcal{E}^2 + M_2^2) + 2M_1^2 \mathcal{P}^2 \cos(2\alpha) + (M_2^2 - s)^2}}{2(\mathcal{E}^2 - \mathcal{P}^2 \cos^2(\alpha))},\end{aligned}$$

$$E_2 = \mathcal{E} - E_1$$

where $s = \mathcal{E}^2 - \mathcal{P}^2$ is the Mandelstam variable introduced earlier

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Evaluation of the relative velocity v_{rel} :

- By definition, this is velocity of one of the particles in the rest frame of the other

$$p_{M_1} = (M_1 c, 0), \quad p_{M_2} = \left(\frac{M_2}{\sqrt{1 - v_{\text{rel}}^2}}, \frac{M_2 \mathbf{v}_{\text{rel}}}{\sqrt{1 - v_{\text{rel}}^2}} \right),$$
$$\Rightarrow (p_{M_1} \cdot p_{M_1}) = \frac{M_1 M_2}{\sqrt{1 - v_{\text{rel}}^2/c^2}} \quad (3)$$

Solving (3), can get invariant relative 4-velocity:

$$v_{\text{rel}} = c \sqrt{1 - \frac{M_1^2 M_2^2}{(p_{M_1} \cdot p_{M_1})^2}}$$
$$p_{M_1} \cdot p_{M_1} = \frac{(p_{M_1} + p_{M_2})^2 - p_{M_1}^2 - p_{M_2}^2}{2}$$

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$$v_{\text{rel}} = c \sqrt{1 - \frac{M_1^2 M_2^2}{(p_{M_1} \cdot p_{M_1})^2}}, \quad p_{M_1} \cdot p_{M_1} = \frac{(p_{M_1} + p_{M_2})^2 - p_{M_1}^2 - p_{M_2}^2}{2}$$

Due to 4-momentum conservation, $p_{M_1} + p_{M_2} = k_1 + k_2$, and since $(k_1 + k_2)^2 = s$ can get that

$$v_{\text{rel}} = c \sqrt{1 - \frac{4 M_1^2 M_2^2}{(4 E_{k_1} E_{k_2} - M_1^2 - M_2^2)^2}}$$

Minimal energy of the process might be found from the condition $v_{\text{rel}} = 0$,

$$s_{\text{min}} \equiv (4 E_{k_1} E_{k_2})_{\text{min}} = (M_1 + M_2)^2 \quad \text{for } E_{k_1} = E_{k_2} \quad \text{get} \quad E_{k_1} = E_{k_2} = \frac{M_1 + M_2}{2}$$

Problem 1 - comment on C-frame

Some students tried to work in C-frame ("CM"-frame):

- redundant, too sophisticated
- just one student made it correctly

I provide what should be done if You like that frame:

$$\text{lab : } k_1^\mu = (E_{k_1}, E_{k_1} \hat{z}), k_2^\mu = (E_{k_2}, -E_{k_2} \hat{z}), E_{k_a} = \sqrt{k_a^2}, a = 1, 2 \quad (1)$$

We need to make Lorentz boost in direction of z with velocity V to get there.

Let's find velocity V needed for this:

- We expect that photon momenta $\tilde{k}_{1,2}$ are equal by magnitude and point in opposite direction, so

$$\tilde{k}_1^\mu = (\tilde{E}_{k_1}, \tilde{E}_{k_1} \hat{z}), k_2^\mu = (\tilde{E}_{k_2}, -\tilde{E}_{k_2} \hat{z}), \quad \tilde{E}_{k_1} \equiv \tilde{E}_{k_2}$$

$$\tilde{E}_{k_1} \equiv \tilde{E}_{k_2} \Rightarrow E_{k_1} \frac{1+V}{\sqrt{1-V^2}} = E_{k_2} \frac{1-V}{\sqrt{1-V^2}} \Rightarrow V = \frac{E_{k_1} - E_{k_2}}{E_{k_1} + E_{k_2}}$$

Problem 1 - comment on C-frame [cont'd]

- We may repeat verbatim what appears in previous slides, replacing $E_{k_1}, E_{k_2} \rightarrow \tilde{E}_{k_1}, k_2 \rightarrow -k_1$
- allows to get some simplifications ...
- expression for meson momenta:

$$p_{M_1}^\mu = (\tilde{E}_{M_1}, \mathbf{p}_1), p_{\pi^-}^\mu = (\tilde{E}_{M_2}, -\mathbf{p}_1), \quad \tilde{E}_{M_a} = \sqrt{\mathbf{p}_1^2 + M_a^2}, \quad a = 1, 2 \quad (2)$$

However,

- angle $\tilde{\alpha}$ in C-frame does not coincide with angle α in lab frame
- meson energies $\tilde{E}_{M_1}, \tilde{E}_{M_2}$ do not coincide with lab frame energies
- ... should use Einstein's formulas to transform components of momentum back to lab frame

Problem 2 - part 1

- ① The lagrangian of the nonrelativistic particle of mass m moving in the external 3D field is given by

$$L = e^{\gamma t} \left(\frac{m \dot{r}^2}{2} - \frac{\kappa r^2}{2} \right), \quad \kappa, \gamma = \text{const}, \quad \kappa < 0, \quad \gamma > 0. \quad (1)$$

- **(3 points)** Write out the Euler-Lagrange equations for this system and compare them with equations for $\gamma = 0$ (spherical harmonic oscillator). How do You expect the new term $\sim \gamma$ will affect the trajectories?

Problem 2 - part 1

Solution: Working in Cartesian coordinates, can get immediately

$$0 = \frac{\partial L}{\partial r_a} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{r}_a} \right) = -e^{\gamma t} (m\ddot{r}_a + \gamma m\dot{r}_a + \kappa r_a) = 0$$

This is equation for damped oscillator in 3D, parameter γ controls damping:

General solution :

$$r_a = A_a e^{-\gamma t/2} \cos \left(t \sqrt{\frac{\kappa}{m} + \left(\frac{\gamma}{2} \right)^2} + \varphi_a \right)$$

Constants A_a , φ_a should be fixed from initial conditions

–For $\gamma = 0$ have oscillations with constant amplitude, for $\gamma \neq 0$ the amplitude of oscillations gradually decreases

Problem 2 - part 2

- Identify continuous symmetries of this system and the integrals of motion associated with them due to Noether's theorem. Analyze if there are continuous symmetries of type

$$t \rightarrow t + \alpha, \quad r \rightarrow r\beta, \quad \alpha, \beta = \text{const}$$

which do not change the action of the system. Demonstrate that all the integrals of motion are conserved using equations of motion.

- ▶ We have seen in lectures that for time-independent central field, the particle's trajectory always remains in the plane (so-called scattering plane). Analyze if this property remains valid for the system described by lagrangian (1). Analyze if the second law of Kepler remains valid for this system.

Problem 2 - part 2 (solution)

- ① The lagrangian has spherical symmetry, i.e. rotations around arbitrary vector $\mathbf{n}$,

$$\mathbf{r} \rightarrow \mathbf{r} + \phi \underbrace{\mathbf{n} \times \mathbf{r}}_{\equiv \boldsymbol{\gamma}}, \quad \phi \ll 1, \quad \phi = \text{const}$$

Repeating derivation of angular momentum conservation from Lecture 6 (page 23), we may get that there is a conserving quantity

$$Q = \mathbf{p} \cdot \boldsymbol{\gamma} = \sum_a \frac{\partial L}{\partial \dot{q}_a} \gamma^a = \mathcal{M} \cdot \mathbf{n} = e^{\gamma t} \mathbf{n} \cdot [\mathbf{r} \times m\mathbf{v}], \quad \mathbf{p} = m\mathbf{v}e^{\gamma t}$$

i.e. the “classical” angular momentum $\mathbf{M} = \mathbf{r} \times m\mathbf{v}$ is not conserved, but instead we have conservation of angular momentum defined as $\mathcal{M} = \mathbf{r} \times \mathbf{p} = e^{\gamma t} \mathbf{M} = \text{const.}$ The variable $\mathbf{M}$ decreases exponentially as expected,

$$\mathbf{M}(t) = \mathcal{M}e^{-\gamma t}$$

Problem 2 - part 2 (solution)

- ① The force acting on the particle

$$m\ddot{\mathbf{r}}_a = -\gamma\dot{\mathbf{r}}_a - \kappa\mathbf{r}_a$$

The force always remains in the same plane as vectors $\mathbf{r}$, $\mathbf{v}$, so the trajectory will remain in this plane forever (as in $\gamma = 0$ case):

Second law of Kepler **is NOT valid now**

- ① Original formulation for $\gamma = 0$:

A line segment joining a planet and the Sun sweeps out equal areas during equal intervals of time

Proof:

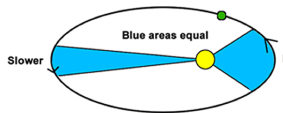
Based on conservation on angular momentum: $M_z \equiv$

$$p_\varphi = mr^2 \frac{d\varphi}{dt} = 2m \frac{dA}{dt} = \text{const}$$

- ② For $\gamma \neq 0$ situation changes:

$$M_z(t) \equiv 2m \frac{dA}{dt} = \mathcal{M}_z e^{-\gamma t} \neq \text{const}, \quad A(t) = -\frac{\mathcal{M}_z}{2m\gamma} e^{-\gamma t}$$

i.e. the area which sweeps out a line segment decreases as a function of time



Problem 2 - part 2 (solution)

$$S = \int L dt, \quad L = e^{\gamma t} \left(\frac{m \dot{\mathbf{r}}^2}{2} - \frac{\kappa r^2}{2} \right), \quad \kappa, \gamma = \text{const}, \quad \kappa < 0, \quad \gamma >$$

Can observe that the action is invariant under

$$t \rightarrow t + \alpha, \quad \mathbf{r} \rightarrow \beta \mathbf{r}, \quad \beta = e^{-\gamma \alpha/2}, \quad \alpha = \text{const}$$

To find Noether invariant, should assume $\alpha = \alpha(t) \ll 1$ and pick the terms $\sim \dot{\alpha}$ (the terms $\sim \alpha$ cancel)

–Caution with differentials:

$$dt \rightarrow dt(1 + \dot{\alpha}), \quad \dot{\mathbf{r}} \equiv \frac{d\mathbf{r}}{dt} \rightarrow \frac{\left(\dot{\mathbf{r}} - \frac{\gamma \dot{\alpha}}{2} \mathbf{r} \right)}{1 + \dot{\alpha}} + \mathcal{O}(\alpha)$$

$$\delta S = \dots = - \int dt \, \dot{\alpha} e^{\gamma t} \left[\frac{m}{2} (\dot{\mathbf{r}}^2 + \gamma \dot{\mathbf{r}} \cdot \mathbf{r}) + \frac{\kappa r^2}{2} \right] + \mathcal{O}(\alpha)$$

Problem 2 - part 2 (solution)

Integral of motion :

$$Q = e^{\gamma t} \left[\frac{m}{2} (\dot{\mathbf{r}}^2 + \gamma \dot{\mathbf{r}} \cdot \mathbf{r}) + \frac{\kappa r^2}{2} \right]$$

For $\gamma = 0$ this is just (minus) energy; for $\gamma \neq 0$ energy is not conserved.
Explicit check that $Q = \text{const}$:

$$\frac{dQ}{dt} = -\frac{1}{2} e^{\gamma t} (2\dot{\mathbf{r}} + \gamma \mathbf{r}) \cdot (m(\ddot{\mathbf{r}}(t) + \gamma \dot{\mathbf{r}}(t)) + \kappa \mathbf{r}(t))$$

For trajectories which are solutions of the equations of motion

$$m\ddot{\mathbf{r}}_a = -\gamma \dot{\mathbf{r}}_a - \kappa \mathbf{r}_a \Rightarrow dQ/dt = 0$$

General solution :

$$\mathbf{r}_a = A_a e^{-\gamma t/2m} \cos \left(t \sqrt{\frac{\kappa}{m} - \left(\frac{\gamma}{2m} \right)^2} + \varphi_a \right)$$

$$\Rightarrow Q = \frac{m}{2} \mathbf{A}^2 \left(\frac{\kappa}{m} - \left(\frac{\gamma}{2m} \right)^2 \right) = \text{const}$$

Problem 2 - parts 3-5

- Generalized momenta and hamiltonian

$$p_a = \frac{\partial L}{\partial \dot{r}_a} = m \dot{r}_a e^{\gamma t}, \quad H = p_a \dot{r}_a - L = \left(\frac{\mathbf{p}^2}{2m} e^{-\gamma t} + \frac{\kappa r^2}{2} e^{\gamma t} \right)$$

- Canonical equations of motion:

$$\dot{r}_a = \frac{\partial H}{\partial p_a} = \frac{p_a}{m} e^{-\gamma t}, \quad \dot{p}_a = -\frac{\partial H}{\partial r_a} = -\kappa r_a e^{\gamma t}$$

If we substitute $p_a \rightarrow m \dot{r}_a e^{\gamma t}$, get immediately

$$m(\ddot{r}_a + \gamma \dot{r}_a) e^{\gamma t} = -\kappa r_a e^{\gamma t}$$

in agreement with Euler-Lagrange.

- The Hamilton-Jacobi equation is

$$\frac{\partial S}{\partial t} + \frac{(\nabla S)^2}{2m} e^{-\gamma t} + \frac{\kappa r^2}{2} e^{\gamma t} = 0.$$

Problem 2 - [part 2++]

Integrals of motion (symmetry generators):

- The invariant Q which we found earlier may be rewritten as

$$Q = e^{\gamma t} \left[\frac{m}{2} (\dot{\mathbf{r}}^2 + \gamma \dot{\mathbf{r}} \cdot \mathbf{r}) + \frac{\kappa r^2}{2} \right] = \left[\left(\frac{\mathbf{p}^2}{2m} e^{-\gamma t} + \frac{\gamma}{2} \mathbf{p} \cdot \mathbf{r} \right) + \frac{\kappa r^2}{2} e^{\gamma t} \right]$$

- Components of vector $\mathcal{M}_a = [\mathbf{r} \times m e^{\gamma t} \dot{\mathbf{r}}] = \mathbf{r} \times \mathbf{p}$
- Poisson theorem: if f and g are integrals of motion, then $h = [f, g]$ is also integral of motion
 - ▶ Can use to recover “new” integrals of motion
 - ▶ Straightforward evaluation of Poisson brackets:

$$[M_a, M_b] = -\epsilon_{abc} M_c, \quad [Q, M_a] = \dots = 0$$

there is no new invariants, algebra is closed.

Problem 2 - [final comment]

We could also replace variable

$$\mathbf{r}(t) \rightarrow e^{-\gamma t/2} \mathbf{s}(t)$$

in order to eliminate $e^{\gamma t}$ in lagrangian, so we would get:

$$L = \frac{m}{2} \dot{\mathbf{s}}^2 - \frac{m\gamma}{2} \mathbf{s}(t) \cdot \dot{\mathbf{s}}(t) - \frac{1}{2} \left(\kappa - \frac{\gamma^2}{4m} \right) \mathbf{s}^2$$

- No explicit dependence on time (only coordinates and velocity), so “energy” $\mathcal{E}$ is conserved, and coincides with Q (!)

$$\mathcal{E} = \sum_a \dot{s}_a \frac{\partial L}{\partial \dot{s}_a} - L = \frac{m}{2} \dot{\mathbf{s}}^2 + \frac{1}{2} \left(\kappa - \frac{\gamma^2}{4m} \right) \mathbf{s}^2$$

$$Q = -e^{\gamma t} \left[\frac{m}{2} (\dot{\mathbf{r}}^2 + \gamma \dot{\mathbf{r}} \cdot \mathbf{r}) + \frac{\kappa r^2}{2} \right] = -\frac{m}{2} \dot{\mathbf{s}}^2 - \frac{1}{2} \left(\kappa - \frac{\gamma^2}{4m} \right) \mathbf{s}^2 = -\mathcal{E}$$

Can see that analysis simplifies drastically (!)

Problem 3

- 1 The transformation of coordinates and momenta has a form

$$p = Q^\alpha \cos(\beta P), \quad q = Q^\alpha \sin(\beta P), \quad \alpha, \beta = \text{const}$$

- ▶ **(5 points)** Find the values of constants α, β for which the transformation is canonical. For these values of α, β , find explicitly the generating function F which corresponds to the transformation.
- ▶ **(5 points)** Apply the transformation found in the previous item to the hamiltonian of harmonic oscillator $H(p, q) = \frac{p^2}{2} + \frac{q^2}{2}$ and find the hamiltonian $\tilde{H}(P, Q)$ in terms of the new variables (P, Q) . Write out the canonical equations in terms of these new variables and explain your result.

Problem 3 - solution

Straightforward evaluation of Poisson bracket yields

$$[p, q]_{P, Q} = -\alpha\beta Q^{2\alpha-1}$$

We know that for canonical transformation right-hand side MUST be $= 1 = 1 \cdot Q^0$, so

$$-\alpha\beta = 1, \quad 2\alpha - 1 = 0 \quad \Rightarrow \quad \alpha = \frac{1}{2}, \quad \beta = -2$$

$$p = \sqrt{Q} \cos(2P), \quad q = -\sqrt{Q} \sin(2P),$$

Let's find explicitly the generating function. There are 4 different types of generating functions. Let's find explicitly $F_1(q, Q)$:

$$P = -\frac{\partial F}{\partial Q} = \sqrt{q - Q^2}, \quad p = \frac{\partial F}{\partial q} = \frac{1}{2} \arccos \frac{\sqrt{q - Q^2}}{\sqrt{q}}$$

$$F(q, Q) = -\frac{Q}{2} \sqrt{q - Q^2} - \frac{q}{2} \arctan \frac{Q}{\sqrt{q - Q^2}}$$

Problem 3 - solution

Hamiltonian after transformation to variables Q, P :

$$H(p, q) = \frac{p^2}{2} + \frac{q^2}{2} \Rightarrow \tilde{H} = \frac{Q}{2}$$

The canonical equations:

$$\dot{P} = -\frac{\partial H}{\partial Q} = -\frac{1}{2}, \quad \dot{Q} = \frac{\partial H}{\partial P} = 0$$

$$Q = \text{const}, \quad P = -\frac{t}{2} + P_0$$

In Lecture 15 we introduced “canonical” action-angle variables (I, w) and applied them to harmonic oscillator.

- “Action” I is integral of motion (energy)
- “Angle” w is cyclic variable
- The variables P, Q of this problem after change of the notations $\{P \rightarrow -w, Q \rightarrow I\}$ coincide with those action-angle variables, that’s why we got similar results